

CS6440: Introduction to Health Informatics Spr24

Practicum Sprint #5 - Status Check In #3

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Abstract—We introduce *SkinWise*, a web-based diagnostic tool enhancing healthcare providers' capabilities in early skin cancer detection and management, focusing on melanoma. Leveraging advanced machine learning techniques and datasets like Skin Cancer ISIC and FHIR, *SkinWise* offers innovative solutions for accurate cancer classification and management.

1 ACCOMPLISHMENTS

In alignment with the tasks outlined in our Sprint #4, we kept working on the planned tasks within the specified timeframe. Table 1 shows our achievements, indicating the primary contributors within our team. We have regular group meetings for internal discussions and keep updating our *SkinWise* [GitHub](#) repository to maintain updates and version control. We successfully linked our web application to the FHIR server hosted at [HAPI FHIR Test](#). We also updated our UI with the functionalities as planned. For example, Figure 1 shows a screenshot of our updated UI with an integration of the skin disease classification model. Figure 2 and Figure 3 demonstrate the functionality for health providers to create, delete, and edit patient data using the CRUD interface. Overall, as we approach the final submission deadline, we are refining our application to ensure it meets all requirements and is prepared for submission.

Project Tasks / Activities	Group Member
Ensure smooth functionality of CRUD interface operations, including Create, Delete, etc.	Jun
Ensure seamless functionality of our chosen image classification model within our web app	Zhiqiu, Zifeng
Test the deployment of our web app (locally and also on AWS)	Jun, Zhiqiu, Zifeng
Update our User Interface	Jun, Zhiqiu
Debug the codebase for our web app	Jun
Ensure smooth local deployment of the CRUD interface for Patient Data Management	Jun, Zhiqiu, Zifeng
Ensure the connection between our web app and the FHIR server	Jun

Table 1—A list of accomplishments

2 CHALLENGES

In the following, we describe the primary challenges encountered thus far during this sprint.

- **Connect the selected skin disease classification model with our web app:** One of our primary challenges lies in seamlessly integrating the skin disease classification model and our web application, particularly in handling data inputs, processing requests efficiently, and displaying accurate results on the interface. We finally ensured its smooth operation and presentation of the model's predicted results in the desired location on our web app interface. Throughout the integration process, we extensively tested it and addressed any bugs that arose.
- **Setup deployment using Docker image:** We successfully set up local deployment using a Docker image, while we faced challenges in finalizing the functionalities of our web application. Therefore, ongoing work includes finalizing web application functionalities and ensuring seamless integration of new features within the Docker environment.
- **Implement CRUD functionalities for FHIR patient data:** We encountered challenges while implementing CRUD functionalities for managing FHIR patient data. Tasks such as creating, deleting, and editing patient records within the FHIR framework required meticulous attention to detail and adherence to FHIR standards. Despite these challenges, we successfully ensured smooth functionality of CRUD interface operations, including Create, Delete, and Edit (see Figure 2 and Figure 3).
- **Integrate Skin Disease Classification Results into FHIR Server:** Another challenge lies in updating the skin disease data and accurately formatting it in JSON for the FHIR server. We have tested this process ourselves and verified the correct JSON format.
- **Bugs:** We need to fix bugs in the codebase to ensure it works for different data as a comprehensive web-based diagnostic tool as planned.

We are currently finalizing the resolution of technical obstacles encountered during SkinWise's deployment as we prepare for the final submission.

3 SPRINT PLANS

Our next check-in is scheduled for the week ahead, aligning closely with our projected timeline (refer to Table 2). As we finalize all tasks, we remain committed to adhering to our implementation plan during the upcoming sprint period.

APPENDIX

Description	URL
Github repo	https://github.gatech.edu/jhuang709/SkinWise.git
AWS EC2 IP	http://13.60.53.97:8501
AWS EC2 DNS	http://ec2-13-60-53-97.eu-north-1.compute.amazonaws.com:8501

 **Data Entry**

 Patients

Data Entry for Patient

Select Birth Day:
1

Select Birth Month:
January

Select Birth Year:
2024

First Name:
Zifeng
Press Enter to submit form

Family Name:
Zhang
Press Enter to submit form

Diagnosis:

Save Data

choose a image file

 Drag and drop file here
Limit 200MB per file

Browse files

 ISIC_0028394.jpg 63.5KB

×

Patient's Name
zzf

Patient's Contact Number
59771519

Doctor's Name
zzz

Doctor's Contact Number
111

Get Answer

Patient Name: zzf Doctor Name: zzz Disease Name : melanocytic nevi Confidence: 1.0

Figure 1—Screenshot of updated UI with an integration of machine learning classification model in our web app

Patient with ID 44663534 deleted successfully!

Patient Information Tracker

 Data Entry

 Patients

Patient Information Visualization

Fetch Latest Patients

Patient ID: 44663549

Edit

Submit

Delete

First Name:

Kun

Family Name:

Kun

Birth Date (YYYY-MM-DD):

2024-01-28

Patient ID: 44663546

Edit

Submit

Delete

First Name:

Jone

Family Name:

Figure 2—Screenshot of updated UI showing how to add and delete patient data on CRUD

Patient with ID 44663549 updated successfully!

Patient Information Tracker

 Data Entry  Patients

Patient Information Visualization

Fetch Latest Patients

Patient ID: 44663549 Edit Submit Delete

First Name:

Family Name:

Birth Date (YYYY-MM-DD):

Patient ID: 44663546 Edit Submit Delete

First Name:

Figure 3—Screenshot of updated UI showing how to edit and update patient data on CRUD

Patient Information Tracker

 Data Entry

 Data Visualization

Data Entry for Patient

Select Birth Month:

January

Select Birth Year:

2024

Patient Information

Name:

Age:

0

Diagnosis:

Skin Treatment Information

Treatment Plan:

Medication:

Follow-up Schedule:

Comment

Enter a comment here ...

Save Data

Figure 4—Screenshot of the app interface hosted on AWS via publicly accessible link

Project Timeline	Project Tasks	Needs	Risks
Jan 8 - Jan 21	Work to form our team.	Identify potential team members with similar project interests and diverse skills and expertise. Establish clear communication channels for team collaboration.	Difficulty in finding suitable team members with the close interests. Communication challenges within the team, e.g., time zones.
Jan 22 - Jan 28	Team formation and project kick off. Submit Practicum sprint # 1.	Formalize team structure and roles. Define project goals and deliverables.	Lack of clarity in project objectives leading to confusion among team members. Initial setbacks in team coordination and workflow.
Jan 29 - Feb 18	Brainstrom team project topic - SkinWise. Assign roles for each team member. Start to explore literature and resources about our project. TA mentor will be assigned.	Research resources and literature relevant to our SkinWise project. Brainstorm the solution. Assign specific roles and responsibilities to team members.	Difficulty in finding relevant and reliable literature/resources. Difficulty in finding relevant health data sources. Potential conflicts arising from role assignments within the team.
Feb 19 - Feb 25	Project planning and design. Identify the focus of SkinWise as a web-based diagnostic tool. Discuss project topic with our TA mentor. Explore architecture design. Submit Practicum sprint # 2.	A project plan outlining the scope, objectives, and timeline of the SkinWise web application. Discussions and consultations with mentor to gather feedback. Explore various architecture design options for the SkinWise. Update our sprint # 2 with mentor's constructive feedback.	Difficulty in finding necessary data sources. Challenges in having interoperability. Shifting project focus leading to scope creep.
Feb 26 - Mar 3	Refine project plan. Discuss more details with mentor. Start data processing from the Skin Cancer ISIC Dataset. Set up project environment.	Detailed project planning and design. Initiate data processing from the Skin Cancer ISIC Dataset. Set up the web app project environment.	Delays in project planning leading to a rushed implementation phase. Technical challenges in processing and handling the dataset. Technical challenges in setting the environment.
Mar 3 - Mar 10	Develop image classification model using deep neural networks. Start the codebase for our web app. Start to implement CRUD Interface for Patient Data Management.	Development of image classification model using deep neural networks. Initial setup of the web app codebase.	Complexity in implementing deep neural networks for image classification. Technical challenges in the web development with connecting with different datasets.
Mar 11 - Mar 17	Call for the patient resources from FHIR. Refine web app codebase. Mentor check-in # 1. Submit Practicum sprint # 3.	Access patient resources from FHIR for integration. Refine web app codebase. Conduct mentor check-in to ensure project alignment.	Delays or issues in obtaining necessary patient resources. Integrating data from FHIR sources can be complex. Relying on FHIR resources introduces dependencies.
Mar 18 - Mar 24	Continue project development. Connect the FHIR skin cancer patient data with the classification model. Web app development.	Continued project development including integration of FHIR data. Progress in web app development.	Difficulties in connecting FHIR patient data with the classification model. Challenges in the development of the web app functionality.
Mar 25 - Mar 31	Testing and validation of the web app. Mentor check-in # 2. Submit Practicum sprint # 4.	Testing and validation of the web app. Final mentor check-in before project submission.	Inadequate testing leading to undiscovered bugs or issues. Insufficient time for addressing feedback from mentor check-in.
Apr 1 - Apr 7	Deployment of the SkinWise web application with CRUD functionality	Deployment of the SkinWise web application using CRUD operations. Examine the deployment online.	Deployment issues causing downtime or functionality failures. issues with data integrity, such as data loss or corruption, especially during data migration or database updates. Incomplete deployment leading to user dissatisfaction.
Apr 8 - Apr 14	Mentor check-in # 3. Project finalization phase. Submit Practicum sprint # 5.	Finalize project details and address any remaining issues. Prepare for project submission.	Online deployment may result in performance degradation, such as slow response times, increased latency, or system crashes, especially under heavy loads. Other challenges in the deployment such as configuration errors, network issues.
Apr 15 - Apr 21	Submit Practicum sprint # 6. Submit any other documents/deliverables if needed.	Submission of the final project deliverables.	Technical difficulties or errors during submission. Incomplete or insufficient project documentations.

Table 2—Implementation plan: project tasks, project timeline, and identified needs/risks